

52) Construct a truth table for the logical operator NOR.

Soln:

p	q	$p \downarrow q$
T	T	F
T	F	F
F	T	F
F	F	T

53) Show that $p \downarrow q$ is logically equivalent to $\neg(p \vee q)$

Soln: Let $p \downarrow q$ be True. By defn. of NOR, p and q are both False. Let $\neg(p \vee q)$ be True. By defn. of negation, $\neg(\neg(p \vee q))$ is False. By double-negation Law, $\neg(\neg(p \vee q)) \equiv (p \vee q)$. $\therefore (p \vee q)$ is False. By defn. of OR, both p, q are False. $\therefore p \downarrow q$ and $\neg(p \vee q)$ are logically equivalent.

54) In this exercise we will show that $\{\downarrow\}$ is a functionally complete collection of logical operators.

a) Show that $p \downarrow p$ is logically equivalent to $\neg p$.

Soln: Let p be True. By defn. of NOR, $p \downarrow p$ is False. By defn. of negation, $\neg p$ is False. Let p be False. By defn. of NOR, $p \downarrow p$ is True. By defn. of negation, $\neg p$ is True. $\therefore p \downarrow p$ and $\neg p$ are logically equivalent.

b) Show that $(p \downarrow q) \downarrow (p \downarrow q)$ is logically equivalent to $p \vee q$.

Soln: Let $(p \downarrow q) \downarrow (p \downarrow q)$ be True. $\therefore$ By defn. of NOR, $(p \downarrow q)$ is False. By defn. of NOR, the following cases arise: ① Both p, q are True ② p is True, q is False ③ p is False, q is True.

Let $(p \vee q)$ be True. By defn. of OR, the following cases arise: ① Both p, q are True ② p is True, q is False ③ p is False, q is True.

c) Conclude from parts (a) and (b) and Exercise 49, that $\{\downarrow\}$ is a functionally complete collection of logical operators.

Soln: From Exercise 49, we know $\neg$ and $\vee$ form a functionally complete collection of logical operators. From parts a and b, we know that we can rewrite any compound proposition of the form $\neg p$ and $(p \vee q)$ using $\{\downarrow\}$ only. $\therefore \{\downarrow\}$ is functionally complete.